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Ordinary Differential Equations

Assignment 5 – MA 417-2025

Topic : Linear System of ODEs, Comparison and Oscillation.

26.09.2025

1.1 Find the exponential operator and the general solution of the system $Y'(t) = AY(t)$ for

(a) $A = \begin{bmatrix} 2 & -2 \\ 1 & 0 \end{bmatrix}$.

(b) $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$.

1.2 Consider the equation $ay'' + by' + cy = 0$, where $a(\neq 0), b$ and c are constants. Transform the given equation into system ($Y' = AY$) and then find the eigenvalues of the matrix A. What is the relation between roots of the characteristic equation $ar^2 + br + c = 0$ and eigenvalues of A?

1.3 Let $A_{n\times n}$ be a real matrix. Show that $\det(e^{tA}) = e^{t(traceA)}$. Conclude that e^{tA} is a nonsingular matrix. Show that for each $t \in \mathbb{R}^n$, $\lim_{Y\to\xi} e^{tA}Y = e^{tA}\xi$ (i.e, $e^{tA} : \mathbb{R}^n \to \mathbb{R}^n$ is continuous). Also show that for each $t \in \mathbb{R}$, the mapping $\xi \mapsto e^{tA}\xi$, from $\mathbb{R}^n$ to $\mathbb{R}^n$, is continuously differentiable.

1.4 Let $\phi(t) = \begin{bmatrix} \sin t & \cos t \\ -\cos t & \sin t \end{bmatrix}$ be a fundamental matrix for the system $Y'(t) = AY(t)$. Find constant matrix $A_{n\times n}$ and compute e^{tA}.

1.5 Consider $Y'(t) = AY(t)$, where A is $n \times n$ matrix. Let $\phi(.)$ be a fundamental matrix of the system. Show that any matrix $\psi(.) \in M_{n\times n}$ is also a fundamental matrix if and only if there exists a nonsingular matrix $C \in M_{n\times n}$ such that $\psi(t) = \phi(t)C$, for all $t \in \mathbb{R}$.

1.6 (a) Let $f(t) = t^a e^{-bt}$, for some positive constants a and b, and let c be a positive number smaller than b. Show that we can find a positive constant K such that $|f(t)| \leq Ke^{-ct}$, $0 \leq t < \infty$.

(b) Let $A = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$, where $\lambda < 0$. Show that there exist positive constants K and α such that $\|e^{tA}X_0\| \leq Ke^{-at}\|X_0\|$, for all $X_0 \in \mathbb{R}^2$. (Hint: Use the expression for e^{tA} and part (a)).

1.7 Let $A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & a \end{bmatrix}$. If $a < 0$, show that for each initial value $X_0 \in \mathbb{R}^3$, there is a positive constant M such that the solution of the IVP satisfies $\|e^{tA}X_0\| \leq M\|X_0\|$ for all $t \geq 0$.

Comparison and Oscillation.

1.8 Show that for $0 \leq a < \infty$, any nontrivial solution of the Hermite equation

$$u''(t) - 2t\,u'(t) + 2a\,u(t) = 0, \qquad t \in \mathbb{R},$$

has at most finitely many zeros.
Hint: Consider the function $v(t) := e^{-t^2/2}\,u(t)$, and determine the differential equation satisfied by $v(t)$.

1.9 Let $q : (0, \infty) \to \mathbb{R}$ be a continuous function such that

$$q(t) \geq \frac{c}{t^2} \quad \text{for some constant } c > \frac{1}{4}.$$

Show that any nontrivial solution u of the differential equation

$$u''(t) + q(t)u(t) = 0$$

has infinitely many zeros in $(0, \infty)$.
Hint. Let u be a solution of $u''(t) + \frac{c}{t^2}u(t) = 0$. Consider the equation satisfied by $v(t) := e^{-t/2}\,u(e^t)$.

1.10 Let $q : [a, b] \to \mathbb{R}$ be a continuous function and let $u, v \in C^1([a, b])$ be twice differentiable on (a, b). Suppose that

$$u(0) = v(0) = 0, \quad u'(0) = v'(0) > 0, \quad u(t) > 0 \text{ for } t \in (a, b),$$

and that

$$\begin{aligned} u''(t) + q(t)u(t) &= 0, \quad t \in [a, b], \\ v''(t) + q(t)v(t) &\geq 0, \quad t \in [a, b]. \end{aligned}$$

Show that

$$v(t) \geq u(t) \quad \text{for all } t \in [a, b].$$